

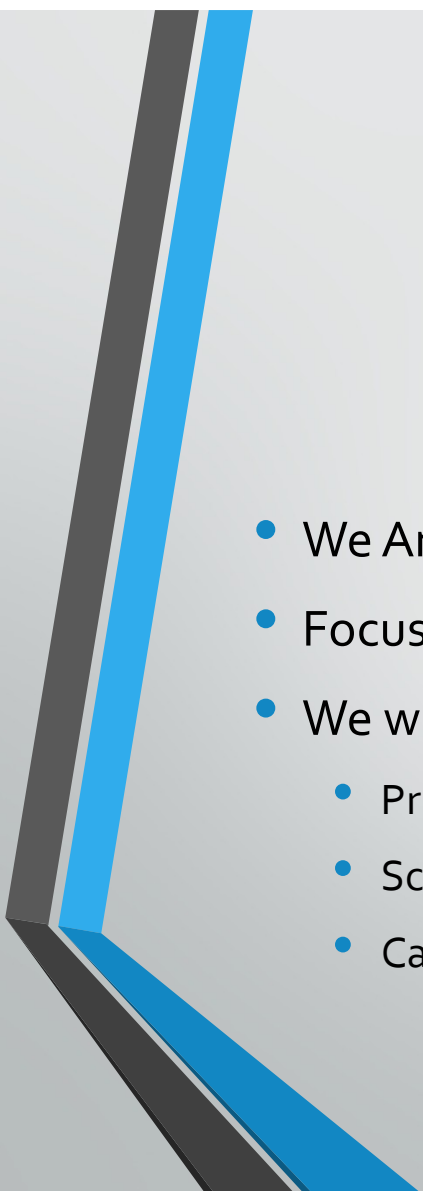


METCS544 – Final Project Presentation

Wireless Network Performance Analysis

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Introduction & Objectives

- We Analyze real world V2X wireless Data
- Focusing on Latency, Throughput, SNR, PER and RSSI Initially.
- We will explore three main questions:
 - Primary factors that affect performance
 - Scenario Impact
 - Can Predictive Models help us in this case efficiently?

Dataset & Preprocessing

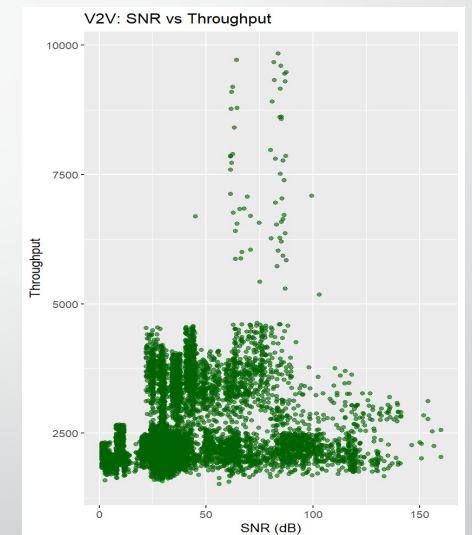
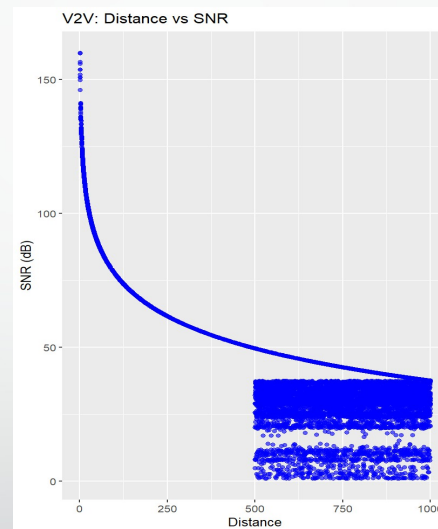
- Dataset: TiHAN-V2X (Hyderabad, India)
- Types: V2V(7 Scenarios) V2I(3 Scenarios)

Preprocessing

- Cleaned Columns and checked for NA or missing value.
- Find out no imputation was needed, Identified outliers
- Changed variable names for easy access

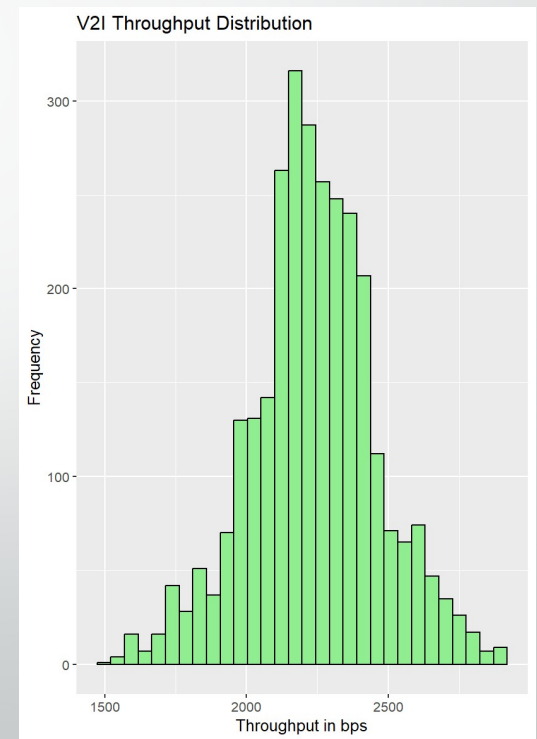
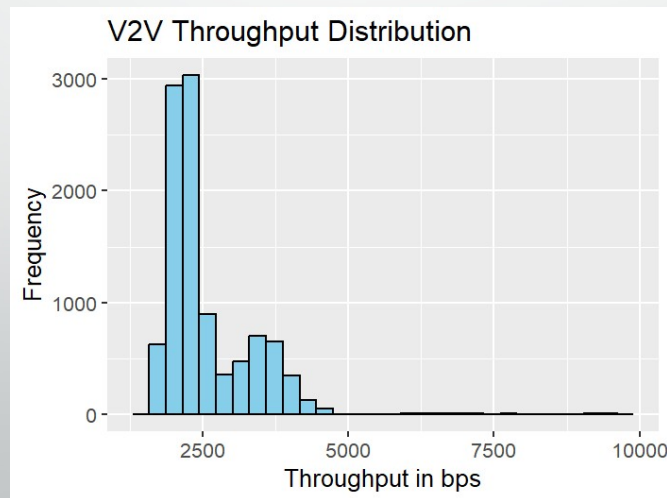
Descriptive Analysis(SOCS & Visuals)

- Throughput = $0.15 * \text{SNR} - 0.02 * \text{Distance}^2$
- Latency: Ultra-low (~0.6 ms)Throughput
- V2V Bimodal (~2-3 kbps & ~6 kbps)
- V2I Unimodal (~2.2 kbps)
- RSSI & SNR: Strongly distance-dependent
- PER: Mostly <5% but increases with distance



Hypothesis Testing

- Latency: 1-sample t-test $\rightarrow$ Latency < 1 ms ($p < 0.0001$)
- Throughput: V2V $>$ V2I throughput ($p < 0.0001$)
- PER: V2V PER $>$ 5% ($p < 0.0001$)



Chi-Square Tests & Anova

- Scenario vs Throughput: ▪ Strong dependency (Cramer's V ≈ 0.935)
- ANOVA: ▪ Significant differences across scenarios ▪ $F \approx 2144$, $p < 0.0001$

```
> chisq.test(table_throughput)

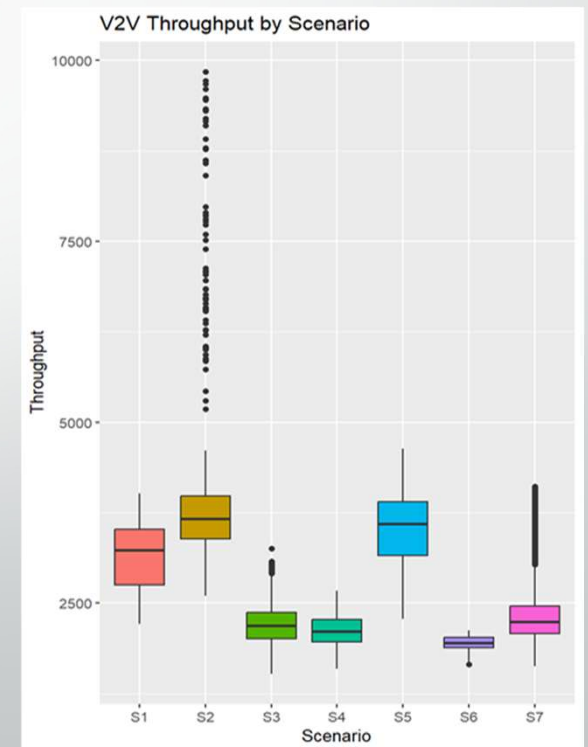
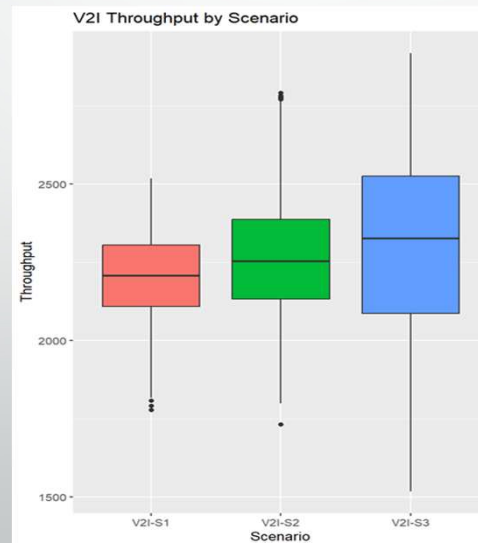
Pearson's Chi-squared test

data: table_throughput
X-squared = 2863.7, df = 6, p-value < 2.2e-16

> summary(anova_latency)
              Df Sum Sq Mean Sq F value    Pr(>F)
factor(scenario) 2  0.151  0.07537    19.2 5.17e-09 ***
Residuals       2953 11.590  0.00392
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> TukeyHSD(anova_latency)
Tukey multiple comparisons of means
 95% family-wise confidence level

Fit: aov(formula = latency_ms_ ~ factor(scenario), data = v2i)

$factor(scenario)~
              diff              lwr              upr              p adj
V2I-S2-V2I-S1 -0.016894902 -0.023315906 -0.010473898 0.0000000
V2I-S3-V2I-S1 -0.008455826 -0.015099676 -0.001811976 0.0080586
V2I-S3-V2I-S2  0.008439076  0.001261512  0.015616639 0.0161739
```



Regression & Decision Tree Models

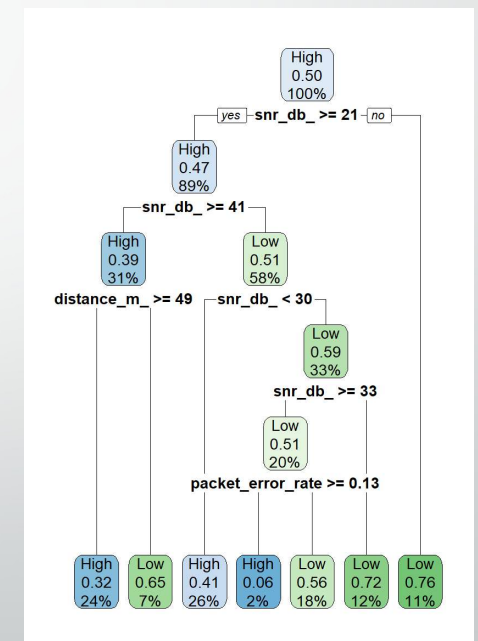
- Regression Analysis:
 - Simple Linear or Regression with Initial metrics R-squared = 0.01-0.55 for V2V
 - Multiple Linear Regression → Adjusted $R^2 \approx 0.869$ with all metrics
- Decision Tree Classification:
 - SNR primary predictor
 - 64% test accuracy

coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	182627.80	14456.49	12.633	< 2e-16 ***
factor(scenario)s2	194.29	20.52	9.468	< 2e-16 ***
factor(scenario)s3	-36.47	15.22	-2.397	0.01656 *
factor(scenario)s4	-56.11	22.20	-2.528	0.01148 *
factor(scenario)s5	21.04	18.11	1.162	0.24541
factor(scenario)s6	-161.91	35.85	-4.516	6.37e-06 ***
factor(scenario)s7	-32.00	15.50	-2.064	0.03907 *
poly(distance_m_, 6)1	-2256.25	307.21	-7.344	2.22e-13 ***
poly(distance_m_, 6)2	1739.64	284.14	6.122	9.56e-10 ***
poly(distance_m_, 6)3	834.65	278.35	2.999	0.00272 **
poly(distance_m_, 6)4	-2406.03	282.82	-8.507	< 2e-16 ***
poly(distance_m_, 6)5	2887.05	281.74	10.247	< 2e-16 ***
poly(distance_m_, 6)6	-2385.63	282.97	-8.431	< 2e-16 ***
transmitted_longitude_deg_	-824.55	337.76	-2.441	0.01466 *
latitude_self_deg_	-1458.95	108.97	-13.389	< 2e-16 ***
longitude_self_deg_	-1113.18	401.24	-2.774	0.00554 **
latency_ms_	-5078.95	33.80	-150.248	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 276.1 on 10235 degrees of freedom
Multiple R-squared: 0.8691, Adjusted R-squared: 0.8689
F-statistic: 4246 on 16 and 10235 DF, p-value: < 2.2e-16



Conclusion

- Latency consistently sub-millisecond
- Throughput depends on message frequency & scenario
- Distance strongly affects SNR and reliability
- Scenario is the dominant performance driver
- Predictive models offer moderate to Good accuracy.
- There are more metrics than SNR, PER, Latency, distance affecting throughput
- Transmitted longitude, self latitude-longitude, latency, non-linear relation of distance and scenario affects throughput.

References

- ANNU ., Pydimarri, P., Rajalakshmi, P., Srikanth, S. V., & Prasad, M. (2024). *TiHAN-V2X: A Comprehensive Dataset for Dynamic C-V2X Communication within the Indian Context*. IEEE DataPort. DOI: 10.21227/f2kd-9g03.
- Ušinskis, V., Makulavičius, M., Petkevičius, S., Dzedzickis, A., & Bučinskas, V. (2021). *Towards Autonomous Driving: Technologies and Data for Vehicles-to-Everything Communication*. Applied Sciences, 11(5), 2195. DOI: 10.3390/app11052195 [pmc.ncbi.nlm.nih.gov](https://pubmed.ncbi.nlm.nih.gov/3411052195/)
- Meena, A. (2025). *Wireless Network Performance Analysis – Final Project Proposal*. (Unpublished project proposal)

THANKYOU